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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|----------------------|------------------------|
| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Session | Stack | Question Information | Level 1   Challenge 48 |
| <p><b>Question description</b></p> <p>Hassan enjoys jumping from one building to the next. However, he merely jumps to the next higher building and stops when there are none accessible. The amount of stamina necessary for a voyage is equal to the xor of all the heights Hassan leaps till he comes to a halt.</p> <p>If heights are [1 2 4], and he starts from 1, goes to 2 stamina required is <math>1 \oplus 2 = 3</math>, then from 2 to 3. Stamina for the entire journey is <math>1 \oplus 2 \oplus 4 = 7</math>. Find the maximum stamina required if can start his journey from any building.</p> <p><b>Constraints</b></p> <p><math>1 \leq N \leq 10^5</math></p> <p><math>1 \leq \text{Height} \leq 10^9</math></p> <p><b>Input</b></p> <p>First line: <math>N</math>, no of buildings.</p> <p>Second line: <math>N</math> integers, defining heights of buildings.</p> <p><b>Output</b></p> <p>Single Integer is the maximum stamina required for any journey.</p> <p><b>Explanation:</b></p> <p>8</p> |         |       |                      |                        |

1 2 3 8 6 4 7 9

Considering the input given by you, 1st starting point is 1. From this building, Hassan will try to find the next higher building that is building 2 here and so on. Likewise he will complete all pass.

Pass 1: 1-> 2->3->8-> 9, XOR = 1

Pass 2: 2->3->8-> 9, XOR = 0

Pass 3: 3->8-> 9, XOR = 2

Pass 4: 8-> 9, XOR = 1

Pass 5: 6->7->9 , XOR = 8

Pass 6: 4->7->9 , XOR = 10

Pass 7: 7-> 9 XOR = 14

Pass 8: 9 XOR = 9

So, the highest XOR value is 14

#### ✓ Logical Test Cases

##### Test Case 1

INPUT (STDIN)

5  
1 2 3 8 6

EXPECTED OUTPUT

11

##### Test Case 2

INPUT (STDIN)

8  
1 2 3 8 6 4 7 9

EXPECTED OUTPUT

14

▼ Mandatory Test Cases

|                                                                          |                                                                          |                                                                           |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <div>Test Case 1</div> <div>KEYWORD</div> <div>arr[1000000]</div>        | <div>Test Case 2</div> <div>KEYWORD</div> <div>for(i=0;i&lt;n;i++)</div> | <div>Test Case 3</div> <div>KEYWORD</div> <div>if(arr[i]&lt;arr[j])</div> |
| <div>Test Case 4</div> <div>KEYWORD</div> <div>st[i]=arr[i]^st[j];</div> |                                                                          |                                                                           |

▼ Complexity Test Cases

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|----------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------|
| <div>Test Case 1</div> <div>CYCLOMATIC COMPLEXITY</div> <div>7</div> | <div>Test Case 2</div> <div>TOKEN COUNT</div> <div>205</div> | <div>Test Case 3</div> <div>NLOC</div> <div>26</div> |
|----------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------|